

SIMATS ENGINEERING

ASSESSMENT TOOL 1: Analytical Problem Solving

COURSE NAME: Cloud Computing and Big Data Analytics **COURSE CODE:** CSA15

COURSE OUTCOME COVERED

CO2: *Analyze virtualization architectures to identify performance and security issues in cloud computing environments.* (BL3)

Dave's Taxonomy: Manipulation (Level 2)
Question Count: 5

Total Marks: 50

Code of Conduct

I V,JAGADISH,V(192521352)_(Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: Jagadish.v_

Assessment Tasks

Analytical Problem Solving

1. Compare Type-1 and Type-2 hypervisors for a cloud datacenter hosting mission-critical applications. Analyze performance, security, and manageability, then justify the better choice.
2. A cloud provider must isolate workloads belonging to finance, healthcare, and student portals on the same hardware. Analyze how virtualization architecture supports isolation and identify two major attack surfaces.
3. Study the following VM host utilization snapshot and recommend corrective actions: CPU 92%, memory 88%, storage I/O latency 35 ms, average VM boot time 170 s. Explain the likely bottlenecks.
4. Analyze how Software Defined Datacenter architecture improves agility compared with a traditional datacenter. Support your answer with compute, storage, and network virtualization considerations.
5. A multi-cloud federation project fails due to identity mismatches between providers. Analyze the issue and propose a secure identity and privacy design using federation concepts.

Analytical Problem Solving Rubric

Criteria	Weightage	Level 4 - Exemplary	Level 3 – Proficient	Level 2 - Developing	Level 1 - Beginning
Problem Interpretation	20%	Interprets all technical requirements accurately	Interprets most requirements correctly	Partial interpretation	Misinterprets the problem
Analytical Reasoning	25%	Strong reasoning with clear cause-effect analysis	Good reasoning with minor gaps	Basic analysis only	Weak or no analysis
Method Selection	20%	Chooses highly appropriate virtualization and security concepts	Chooses mostly suitable concepts	Partially suitable concepts	Inappropriate approach
Solution Quality	20%	Provides feasible and technically sound solution	Reasonable solution with minor issues	Basic solution with limitations	Unclear or invalid solution
Presentation of Analysis	15%	Well-structured and technically precise	Generally clear	Somewhat unclear	Difficult to follow

Total Marks Awarded:

1. Comparison of Type-1 and Type-2 Hypervisors for Mission-Critical Cloud Applications

Feature	Type-1 Hypervisor (Bare-Metal)	Type-2 Hypervisor (Hosted)
Installation	Runs directly on hardware	Runs on top of a host OS
Performance	High performance, low latency	Lower performance due to host OS overhead
Security	More secure with smaller attack surface	Less secure because both host OS and hypervisor can be attacked
Reliability	Highly reliable for enterprise workloads	Depends on host OS stability
Manageability	Centralized management and scalability	Suitable for testing and desktop virtualization
Examples	VMware ESXi, Microsoft Hyper-V, Xen	VMware Workstation, Oracle VirtualBox

Analysis

- **Performance:** Type-1 hypervisors directly access hardware resources, reducing overhead and improving CPU, memory, and I/O efficiency.
- **Security:** Since there is no intermediate operating system, the attack surface is smaller, making Type-1 hypervisors more secure.
- **Manageability:** Enterprise features such as live migration, resource pooling, load balancing, and high availability are better supported in Type-1 hypervisors.

2. Workload Isolation Using Virtualization Architecture

A cloud provider hosting **finance, healthcare, and student portal applications** on the same physical server can use virtualization to create separate **Virtual Machines (VMs)**.

The hypervisor:

1. Allocates dedicated virtual CPUs, memory, storage, and network interfaces to each VM.
2. Prevents direct access between VMs.
3. Enforces resource and security boundaries.
4. Supports multi-tenancy while maintaining isolation.

Example:

- VM1 → Finance Application
- VM2 → Healthcare Application
- VM3 → Student Portal

Even though all VMs share the same hardware, each workload operates independently with its own operating

system and applications.

Two Major Attack Surfaces

1. Hypervisor Attacks

- Attackers exploit vulnerabilities in the hypervisor.
- A successful compromise can affect multiple VMs.
- Example: Privilege escalation or hypervisor takeover.

2. VM Escape Attacks

- An attacker gains access from a guest VM to the host system or another VM.
- This breaks workload isolation and may expose sensitive financial or healthcare data.

3. VM Host Utilization Analysis and Corrective Actions

Given Snapshot

- **CPU Utilization:** 92%
- **Memory Utilization:** 88%
- **Storage I/O Latency:** 35 ms
- **Average VM Boot Time:** 170 s

Likely Bottlenecks

1. CPU Bottleneck

- CPU utilization of 92% indicates the host is heavily loaded.
- VMs may experience scheduling delays and reduced responsiveness.

2. Memory Bottleneck

- Memory utilization of 88% is close to saturation.
- The hypervisor may use memory swapping or ballooning, which degrades performance.

3. Storage Bottleneck

- Storage latency of 35 ms is significantly high for a virtualized environment.
- Slow disk operations increase application response time and VM startup time.

4. Slow VM Provisioning

- VM boot time of 170 seconds suggests delays caused by CPU contention, insufficient memory, and slow storage access.

4. How Software Defined Datacenter (SDDC) Improves Agility

Traditional Datacenter

- Physical servers, storage, and networking are manually configured.
- Provisioning new resources can take days or weeks.
- Scaling requires hardware installation and manual intervention.

Software Defined Datacenter (SDDC)

An SDDC virtualizes and automates compute, storage, and networking resources through software.

Compute Virtualization

- Physical servers are abstracted into virtual machines.
- Workloads can be provisioned, migrated, and scaled quickly.
- Supports automatic resource allocation.

Benefit: Faster deployment and improved utilization.

Storage Virtualization

- Multiple storage devices are pooled into a single logical storage resource.
- Capacity can be allocated dynamically.
- Simplifies backup and disaster recovery.

Benefit: Flexible and efficient storage management.

Network Virtualization

- Logical networks are created independent of physical hardware.
- Virtual switches, routers, and firewalls can be deployed instantly.
- Supports micro-segmentation for security.

5. Multi-Cloud Federation Identity Mismatch Analysis

Problem Analysis

In a multi-cloud federation, users access resources across different cloud providers.

Identity mismatches occur because:

- Different providers use different identity formats.
- User attributes and roles are inconsistent.
- Authentication mechanisms vary across providers.
- Duplicate identities may be created.

These issues cause:

- Access failures.
- Authorization errors.
- Security and compliance risks.

Secure Identity Federation Design

1. Centralized Identity Provider (IdP)

- Use a trusted Identity Provider to authenticate users once.
- All cloud providers trust the same IdP.

2. Single Sign-On (SSO)

- Users authenticate once and access resources across clouds.
- Reduces password management issues.

3. Federation Standards

- Implement SAML, OAuth 2.0, or OpenID Connect.

- Ensures interoperability among providers.

4. Attribute Mapping

- Standardize user roles and attributes across clouds.
- Example:
 - Finance_Admin → Same role in all providers.
 - Student_User → Consistent permissions everywhere.

5. Privacy Protection

- Share only necessary identity attributes.
- Use encrypted tokens and secure communication channels.
- Apply role-based access control (RBAC).